EDS 223 Homework Assignment 1 Revision

Map Making Practise

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Load libraries and data

```
library(tidyverse)
library(stars)
library(here)
library(sf)
library(tmap)
```

```
# Load data
ejscreen <- st_read(here("data", "ejscreen",
"EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb"))
```

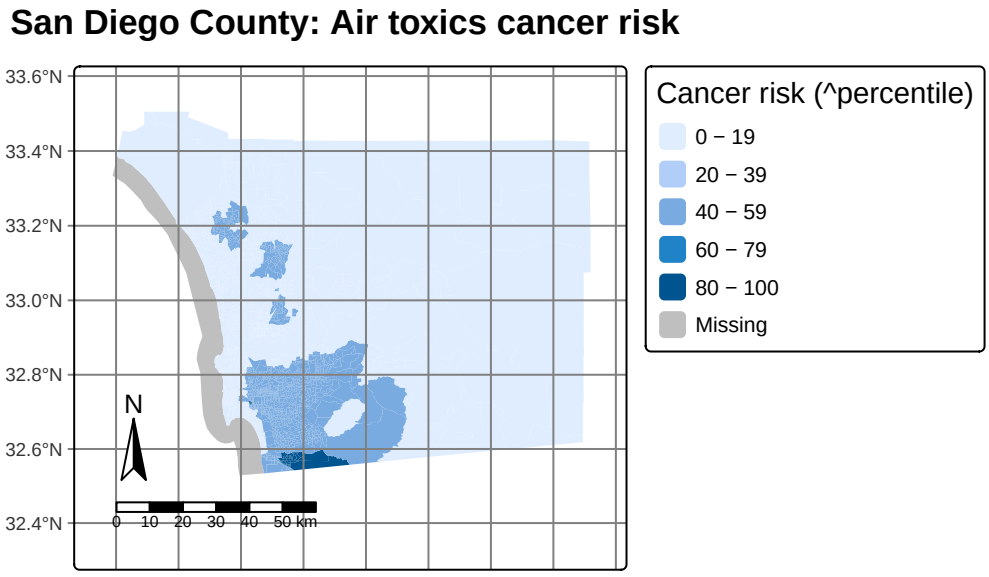
Filtering

```
# subsect for San Diego County
sandiego <- ejscreen |>
    filter(STATE_NAME == "California") |>
    filter(CNTY_NAME == "San Diego County")
```

Mapping

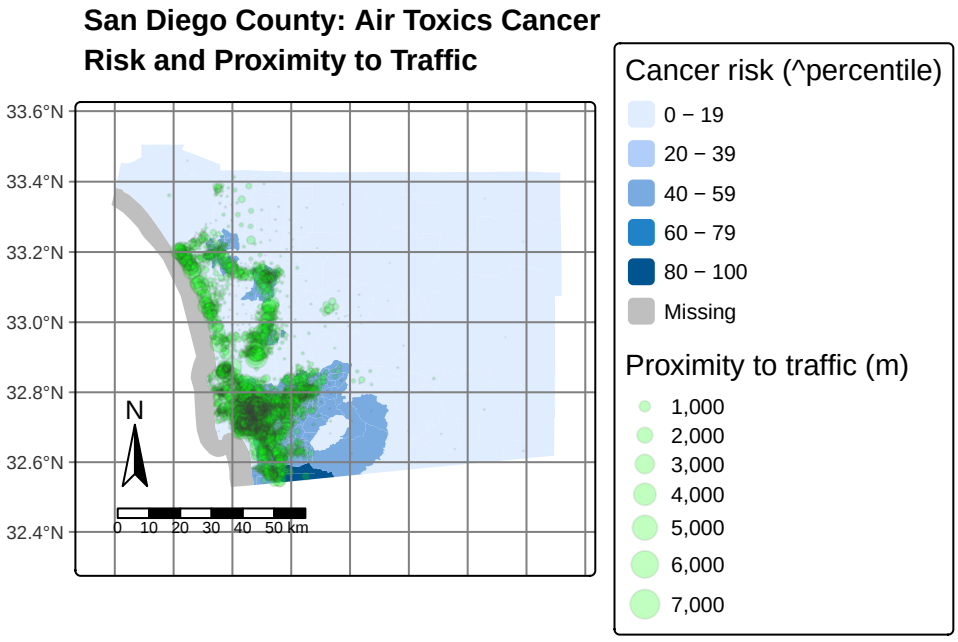
```
tm_shape(sandiego , bbox = c(-117.7, 32.3, -116, 33.6)) +
    tm_polygons(
        "P_CANCER",
```

```
        fill.legend = tm_legend(title = "Cancer risk (^percentile)",
                                position = tm_pos_out("right")),
        col_alpha = 0
    ) +
    tm_title(text = "San Diego County: Air toxics cancer risk",
             size = 1.1,
             fontface = "bold") +
    tm_layout(component.autoscale = FALSE) +
    tm_compass(position = c("bottom","left")) +
    tm_graticules() +
    tm_scalebar(position = c("bottom","left"))
```




```
tm_shape(sandiego , bbox = c(-117.7, 32.3, -116, 33.6)) +
    tm_polygons(
        "P_CANCER",
        fill.legend = tm_legend(title = "Cancer risk (^percentile)",
                                position = tm_pos_out("right"),
                                height = 11),
        col_alpha = 0
    ) +
    tm_symbols(
        size = "PTRAF",
```

```
        fill = "green",
        size.legend = tm_legend(title = "Proximity to traffic (m)",
                                position = tm_pos_out("right"),
                                height = 11),
        fill_alpha = 0.25,
        col_alpha = 0.1
    ) +
    tm_compass(position = c("bottom","left")) +
    tm_graticules() +
    tm_scalebar(position = c("bottom","left")) +
    tm_title("San Diego County: Air Toxics Cancer \nRisk and Proximity to Traffic",
           size = 0.9, fontface = "bold") +
    tm_layout(component.autoscale = FALSE)
```



^According to EJScreen, the air toxics cancer risk indicator was calculated to a statistical percentile. It results in a range from 0 to 100%, with the 35th percentile corresponding to a raw value of 30.

In the original assignment, I wanted to plot the percentitle of cancer, but I picked the wrong column from EJScreen. Therefore, I revised and changed `CANCER` to `P_CANCER` to properly reflect that on both maps.